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IE 3524 Integrated Manufacturing Systems

Analysis of Alkim Kağıt Converting Process

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1. Executive Summary

This report analyzes the converting process of Alkim Kağıt Sanayi ve Ticaret A.Ş., focusing on the Cameron slitter operation in which jumbo paper reels are cut into customer-required widths. The main operational problem is the generation of trim loss due to inefficient cutting pattern selection and additional setup time caused by frequent pattern changes. Since the jumbo reel has a gross width of 350 cm and a usable net width of 348 cm after edge trimming, the way customer widths are combined directly affects material efficiency.

The analysis is based on seven approved production orders involving three customer widths: 57 cm, 64 cm, and 70 cm. In the current system, five cutting patterns are used. These patterns result in 62,371.26 kg of good output and 2,938.74 kg of trim loss, corresponding to a weighted-average trim loss of 4.50%. The observed pattern-level trim loss ranges from 1.72% to 8.05%, indicating that pattern selection is a major driver of waste. The current order sequence also creates six pattern transitions, leading to a total setup time of 105 minutes.

A scenario-based experimental approach supported by Excel Solver was used to test improvement alternatives. The study applied a 2×2 DoE structure with two factors: cutting pattern strategy and order sequencing. The proposed strategy uses Solver-based pattern optimization and grouped order sequencing. The Solver model selected three active patterns and reduced total trim loss to 2,731.40 kg, or 4.20% of raw paper consumption.

Compared with the baseline, the proposed system reduces trim loss by 207.33 kg, corresponding to a 7.06% relative reduction. Setup time decreases from 105 minutes to 45 minutes, and pattern transitions decrease from six to two. Based on the project's 90,000 ton/year capacity assumption, this trim reduction corresponds to an estimated annual saving of approximately 274 tons of paper.

The recommended improvement is to use an Excel Solver-based planning tool for cutting pattern selection and apply pattern-based order sequencing in SAP whenever due-date constraints allow. The results show that measurable improvements can be achieved without major equipment investment. The main conclusion is that better planning before cutting can reduce material waste, decrease setup time, and improve operational consistency in the converting process.

2. Introduction

In recent years, increasing product variety and customer-specific demands have made manufacturing systems more complex, especially in make-to-order (MTO) environments. In these systems, production is directly driven by incoming orders. This creates variability. As a result, planning, scheduling, and resource utilization become more difficult to manage.

The paper manufacturing industry is a good example of this structure. Production generally has two main stages. First, jumbo reels are produced in large-scale paper machines. Then comes the converting stage, where these reels are processed based on customer requirements. Among these operations, slitting stands out as one of the most critical ones, since it directly affects both material usage and overall system efficiency.

In the slitting process, jumbo reels with a fixed width of 350 cm are cut into smaller rolls according to customer demands. However, because order sizes vary, it is almost never possible to fully utilize the reel width. Inevitably, some material is left unused. This is referred to as trim loss or “fire.” In the literature, this situation is commonly defined as a one-dimensional cutting stock problem, where the aim is to meet demand while minimizing waste.

Based on the available production data, waste levels vary depending on the selected cutting patterns. The observed waste ratio ranges between 1.72% and 8.05%, with an average value of around 4.5%. This might seem small at first. But over large production volumes, it becomes quite significant. It also shows that cutting pattern decisions have a clear impact on material efficiency.

Another factor that affects system performance is the setup process. Each pattern change requires a machine adjustment, and during this time production stops. Exact setup times are not included as direct fields in the production data file. Therefore, the analysis uses plant operating practice values: 5 minutes for a same-pattern reel reload and 15 minutes for a full pattern changeover.

The system operates as a single-machine environment, meaning only one reel can be processed at a time. Because of this, the system is quite sensitive to both setup frequency and cutting decisions. So it is not just about minimizing waste. Setup changes also matter, and both need to be considered together.

This study focuses on analyzing the slitting process using real production data. Some parameters are not fully available, so a few assumptions are made where necessary. The goal is to still keep the model realistic.

A scenario-based experimental approach supported by Excel Solver is used to represent the system and test alternative planning strategies. In addition, a Design of Experiments (DoE) framework is applied to evaluate how key factors affect system performance.

Overall, the study aims to reduce material waste, improve machine utilization, and increase production efficiency, while still maintaining the flexibility required in a make-to-order system.

3. Company Overview

3.1. General Information about Alkim Kağıt

Alkim Kağıt Sanayi ve Ticaret A.Ş. is a paper manufacturing company located in the Kemalpaşa Organized Industrial Zone in İzmir, Türkiye. The company operates in the paper and paperboard industry and focuses on producing paper products for office, printing, and commercial use. Its registered facility address is stated as Kemalpaşa Organized Industrial Zone, Kırovası Mevkii, Kemalpaşa/İzmir.

The company's production system can be classified as a process-based manufacturing system, since raw materials such as cellulose, water, and chemicals are transformed through sequential operations into paper reels and then converted into customer-specific products. The main production flow includes pulp preparation, paper forming, drying, winding, slitting, converting, packaging, warehousing, and shipment.

Alkim Kağıt uses several information and control systems to support production and management activities. SAP and S/4HANA are used for areas such as material management and production tracking, while DCS and QCS are used for production monitoring and quality control. These systems support coordination between production, planning, quality control, warehouse management, finance, and other departments.

3.2. Facility Location and Layout

Alkim Kağıt's production facility is located in Kemalpaşa Organized Industrial Zone, Kırovası Mevkii, Kemalpaşa/İzmir. The same location is listed as the company's head office and production facility address in the Public Disclosure Platform records. The official company

contact page also identifies the Kemalpaşa O.S.B. facility as the General Directorate, Factory, and Foreign Trade Sales address.

Figure 1. Location of Alkim Kağıt Kemalpaşa facility in İzmir.



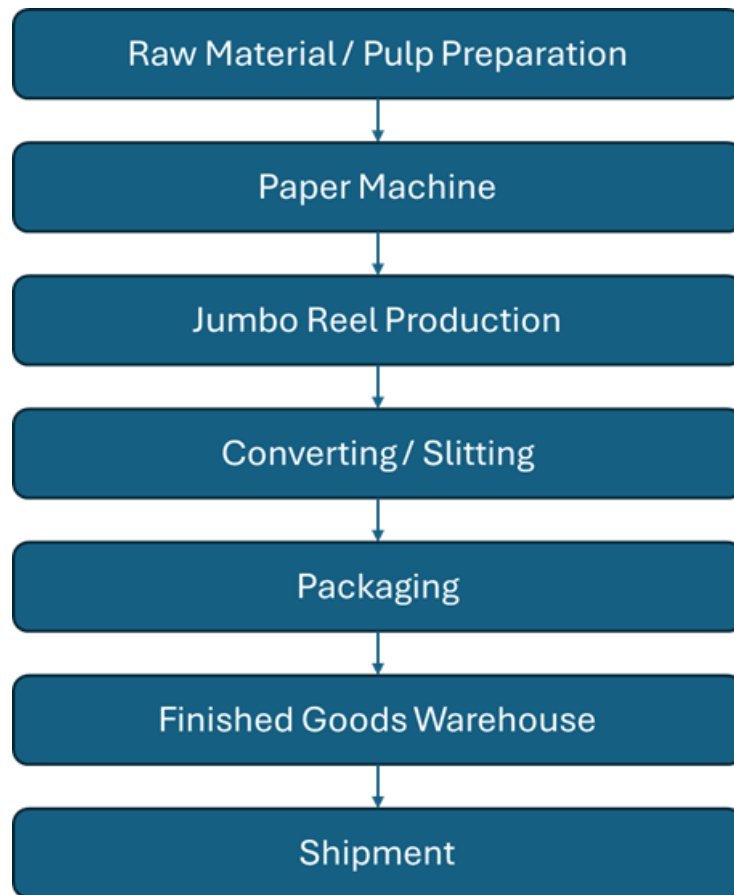
Source: Adapted by the authors using publicly available location information.

From an operations perspective, this location provides advantages for manufacturing and logistics activities. Being located in an organized industrial zone supports access to industrial infrastructure, transportation routes, suppliers, and labor. This is important for Alkim Kağıt because the production system requires continuous inbound material flow, internal material handling, and outbound shipment of finished paper products.

The facility layout can be evaluated as a process-oriented production structure. In the main production area, paper is produced through sequential operations such as pulp preparation, paper forming, pressing, drying, and winding. After jumbo reels are produced, they are transferred to the converting area, where operations such as slitting, cutting, packaging, palletizing, warehousing, and shipment are carried out. Therefore, the converting department acts as a critical link between main paper production and customer-specific final products.

For this project, the layout analysis focuses especially on the material flow from jumbo reel production to the converting/slitting area. Since customer orders require different reel widths and cutting combinations, the position of the slitting operation within the overall facility flow is important for understanding waste generation, setup changes, and production efficiency.

Figure 2. Simplified production and material flow in Alkim Kağıt facility.



Source: Prepared by the authors based on company-provided information and field observations.

3.3. Products and Production Capacity

Alkim Kağıt produces several paper product groups, including offset paper, photocopy paper, coated paper, and specialty papers. The company's Kemalpaşa facility produces paper in the range of 60–220 g/m², including first-grade offset, photocopy, and single- or double-sided coated printing papers. In addition, the company's product range includes label paper, soap packaging paper, cup paper, envelope paper, and ozalit paper. The company has an installed production capacity of 90,000 tons/year, and its offset, photocopy, and industrial paper products are used by printing houses, stationery markets, written press organizations, and industrial facilities.

Table 1. Product groups and production characteristics

Product group	Typical form	Typical use area	Relevant project impact
Offset paper	Reel / sheet	Printing, publishing	Different widths increase slitting complexity
Photocopy paper	A3 / A4 / sheet	Office use	Standardized sizes, high demand
Coated paper	Sheet / reel	Printing and packaging	Quality and surface requirements
Specialty papers	Reel / sheet	Label, cup, envelope, packaging	Product variety increases planning difficulty

Source: Compiled by the authors based on Alkim Kağıt's official facility information and 2024 annual report. The "Relevant project impact" column reflects the authors' manufacturing systems interpretation.

The product variety is relatively high, with approximately 155 product types considered in the available project data. This variety increases the importance of production planning, cutting pattern selection, and converting operations, because different customer orders may require different reel widths, sheet sizes, and packaging forms.

The company's annual production capacity is reported as approximately 90,000 tons for the existing facility. In addition, a second paper machine investment has been announced with the aim of increasing the company's production capacity from 90,000 tons to 210,000 tons after completion. Therefore, for the current project analysis, the existing 90,000-ton capacity should be used as the baseline unless the new investment is specifically included in the project scope.

For this project, the most relevant product group is the paper reel and sheet products processed in the converting stage. Jumbo reels are cut into smaller reels or sheets according to customer orders. Since jumbo reel dimensions do not always match order requirements, material waste and setup time may occur during slitting. Therefore, the product variety and capacity structure directly affect the main problem area of the project.

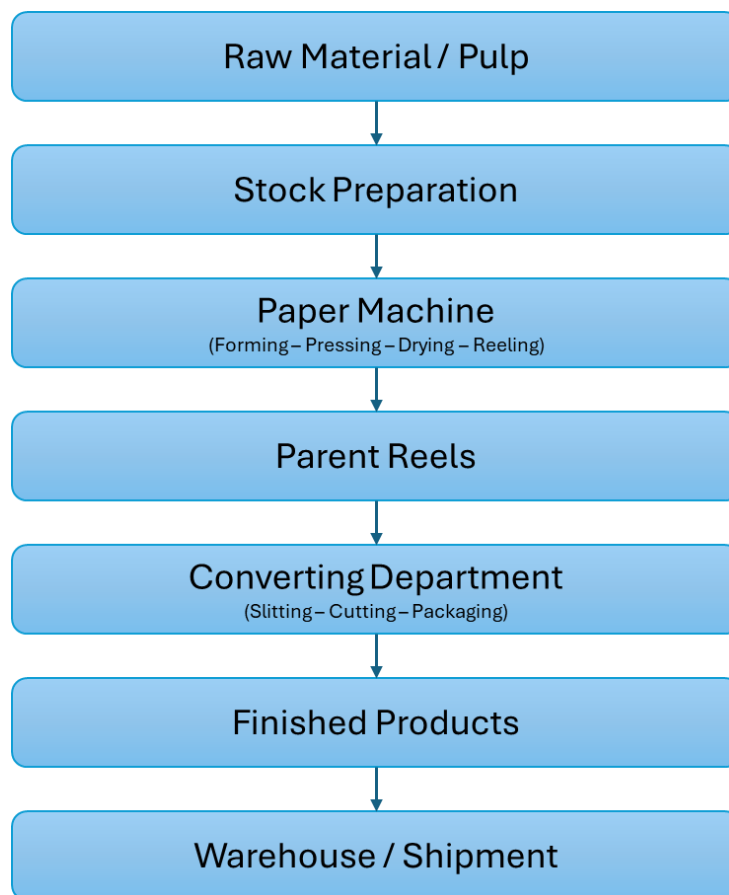
4. Production System Description

4.1. General Paper Production Process

Alkim Kağıt operates as an integrated paper manufacturing facility in which raw material preparation, paper production, finishing, and converting activities are carried out within the

same production system. The production process starts with pulp and chemical preparation, where the required fiber mixture and additives are prepared according to the target paper grade. The prepared stock is then fed into the paper machine, where the paper web is formed, pressed, dried, and wound into large parent reels. Public company information indicates that the existing facility has an annual production capacity of approximately 90,000 tons/year. A second paper machine investment has also been announced to increase total capacity to 210,000 tons/year after completion. Therefore, the current analysis is based on the existing 90,000-ton/year baseline.

Figure 3. General paper production flow at Alkim Kağıt.



Source: Prepared by the authors based on project observations.

After the parent reels are produced, they are transferred to finishing and converting operations. In paper manufacturing, converting generally refers to post-paper-machine operations such as rewinding, trimming, sheeting, and cutting, where large paper reels are transformed into saleable product forms. This stage is especially important for Alkim Kağıt because the company produces different paper types and product dimensions, including offset, photocopy, coated, label, envelope, blueprint, and cup papers in different grammage ranges.

From a production system perspective, the overall process can be considered a combination of continuous production and batch/order-based finishing. The paper machine side operates continuously and focuses on stable quality, high utilization, and process control. In contrast, the converting side is more sensitive to customer orders, product dimensions, cutting plans, packaging requirements, and due dates. Therefore, the interface between paper production and converting is one of the most critical points for production planning and material flow.

4.2. Converting Department

The converting department is the section where large parent reels are transformed into final product forms such as smaller reels, sheets, A3/A4 photocopy paper, and packaged paper products. According to public company information, the facility has eight slitting/cutting machines, and modern converting machines are used to convert paper rolls into smaller product forms. This makes the converting department a key area for the project because product variety, size changes, machine setups, and order priorities directly affect production efficiency.

Table 2. Main converting operations and their project relevance

Converting operation	Main input	Main output	Project relevance
Slitting	Parent reel	Narrower reels	Different customer widths increase cutting-plan complexity
Sheeting / cutting	Reel paper	Sheets or reams	Product size changes may create setup and sequencing issues
Ream packaging	Cut sheets	Packaged photocopy or sheet products	Packaging speed and material availability affect output flow
Roll packaging	Finished reels	Wrapped and labeled reels	Important for product protection, traceability, and shipment readiness
Labeling / identification	Finished product	Identified product units	Supports tracking, warehouse control, and order accuracy

In this project, the converting department should be examined as a production planning and material handling problem rather than only as a set of machines. Since different products may require different widths, grammages, package types, and customer specifications, the same parent reel can be converted into different finished products depending on demand. This creates

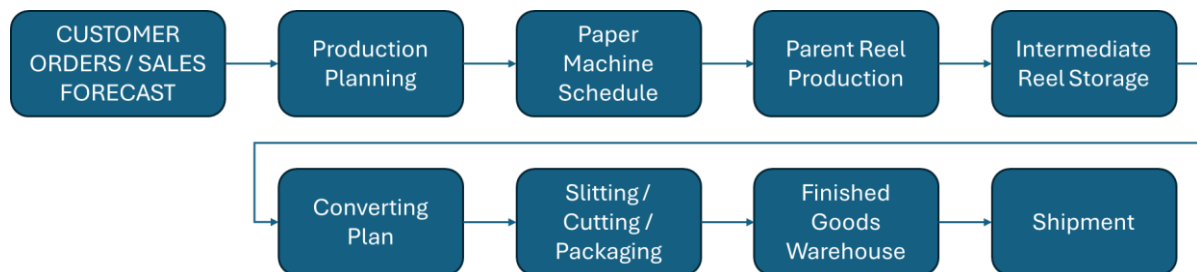
a need for effective order sequencing, reel allocation, cutting optimization, and coordination between production, converting, quality control, and warehouse operations.

A practical issue in this department is that machine utilization is not determined only by processing time. Setup times, reel changes, cutting width adjustments, packaging material changes, and waiting times between operations may also affect the effective capacity of the department. Therefore, the converting department is a suitable area for analyzing bottlenecks, material flow delays, layout-related movements, and improvement opportunities.

4.3. Material and Information Flow

The material flow begins with raw materials and additives entering the stock preparation area and continues through the paper machines, where parent reels are produced. These reels are then moved to intermediate storage or directly to converting, depending on the production plan and order requirements. After converting, the finished products are packaged, labeled, stored, and prepared for shipment. Since the facility produces multiple paper groups and product forms, material flow is not limited to a single standardized route; instead, different products may follow different converting and packaging paths.

Figure 4. Simplified material and information flow in the production system.



Information flow is also essential in this system. Customer orders, product specifications, due dates, grammage, dimensions, and packaging requirements are transferred to production planning. The production plan determines which paper grades will be produced on the paper machines and how the produced reels will be assigned to converting operations. In the converting stage, information such as cutting dimensions, order quantities, packaging type, and product labels must be accurately communicated to prevent wrong production, rework, or delivery delays.

The relationship between material flow and information flow is especially important for the project. If production data, order priorities, machine availability, and inventory levels are not synchronized, the system may experience waiting reels, unnecessary material movement, low

machine utilization, or late deliveries. For this reason, the converting department should be analyzed not only physically, but also in terms of information accuracy, planning logic, and coordination between departments.

4.4. Automation and Control Systems

Automation plays an important role in Alkim Kağıt's production system, especially in the paper machine area. Public company information states that production is supported by DCS and QCS systems. DCS is used to control the main process variables in production, while QCS is used to monitor and control product quality parameters. The company also states that paper production is carried out through computer-assisted systems and that all machines are controlled by PLCs connected to DCS and QCS systems.

In the paper machine area, automation is necessary because the process is continuous and sensitive to variations in speed, moisture, thickness, basis weight, temperature, and web tension. Even small deviations can affect product quality and cause waste. Therefore, automation and control systems support process stability, reduce manual intervention, and help maintain consistent product quality.

In the converting department, automation is mainly related to slitting, cutting, packaging, labeling, and product handling operations. Public information also mentions modern converting machines as well as automatic cutting and roll-packaging systems. These systems are important because they reduce manual workload, increase repeatability, and support faster transformation of parent reels into final product forms.

Table 3. Automation and control functions in the production system

System technology	/ Area of use	Main function	Expected benefit
DCS	Paper machine process	Process monitoring and control	Stable continuous production
QCS	Paper quality control	Quality measurement and adjustment	Consistent paper properties
PLC control	Machines and equipment	Machine-level automation	Reduced manual intervention
Automatic cutting systems	Converting	Sheet or reel cutting	Higher repeatability and speed

System technology	/ Area of use	Main function	Expected benefit
Roll packaging systems	Converting / finishing	Product wrapping and protection	Improved shipment readiness
Barcode labeling logic	Finished products	Product identification	Better traceability and warehouse control

Overall, the production system can be described as a partially integrated and automated manufacturing system. The paper machine side has a high level of process automation, while the converting department combines automated equipment with planning-dependent operations. For this reason, the main improvement potential is expected to be related not only to machine technology, but also to production planning, material flow, scheduling, data usage, and coordination between departments.

5. Selected Process: Slitting Operation

5.1. Description of the Slitting Process

The slitting operation is the converting stage in which jumbo paper reels produced on the paper machine are longitudinally cut into narrower customer-sized rolls. At the Alkim Kağıt Kemalpaşa facility, jumbo reels exit the paper machine with a gross width of 350 cm. After the two outer edge trims of approximately 1 cm each, the net usable width on the slitter is 348 cm. Customer orders, in contrast, arrive in widths between 40 cm and 120 cm and in different grammage classes (typically 70, 80, and 90 g/m²). Because each jumbo reel must be cut into a combination of narrower rolls, the planner is required to design a cutting pattern that simultaneously satisfies customer demand and minimizes the unused edge waste, referred to in the industry as trim loss.

A cutting pattern is denoted in the planning system in the form $w_1-w_2-\dots-w_n$, where each value represents the width of a finished roll obtained from the same jumbo. For instance, the pattern 70-70-70-70-57 places four rolls of 70 cm and one roll of 57 cm side-by-side on a single jumbo, summing to 337 cm of used width and producing 11 cm of trim loss per jumbo. Each pattern change requires the slitter blades and roll cores to be repositioned, which generates non-productive setup time and reduces the effective capacity of the machine.

Within the broader operations management literature, the slitting problem is recognized as a one-dimensional cutting stock problem (1D-CSP). The objective is to satisfy a list of demand items of various widths using as few standard-width stock rolls as possible. In a make-to-order paper plant the problem is further complicated by the fact that orders arrive over time, due dates differ across customers, and the number of distinct cutting patterns is itself an indirect cost driver because each new pattern adds a setup.

5.2. Machines and Equipment Used

Public information about the Alkim Kağıt Kemalpaşa facility states that the converting department is equipped with eight slitting and cutting machines. These machines do not perform identical functions: some operate as primary slitters that convert jumbo reels into narrower customer rolls, while others perform secondary operations such as sheeting and ream cutting for A4 and similar finished formats. Because the production records made available for this study come exclusively from the primary slitting machine known internally as the Cameron slitter, the analysis presented in this report is restricted to that machine. Trim losses, setup times, and pattern decisions reported here therefore refer to the Cameron slitter only. The downstream sheet-cutting operations are excluded from the scope because no measurement data were available for them.

The Cameron slitter receives a jumbo reel from the paper-machine winder, unwinds it under controlled tension, applies the planned set of slitter knives, and rewinds the cut strips into separate finished rolls onto smaller cores. The decision rules used to assign jumbos to orders depend on planner experience, the SAP/S-4HANA-driven order list, and customer due dates. There is no commercial cutting-stock optimizer currently embedded in the workflow, which is the practical motivation for the experimental study described in Sections 6 and 7. The operating parameters of the machine that are directly available from the production data are summarized in Table 4.

Table 4. Slitting line parameters used in the analysis (Cameron slitter).

Parameter	Value	Source
Jumbo reel gross width	350 cm	Production records
Edge trim per side	1 cm	Derived from data

Parameter	Value	Source
Net usable jumbo width	348 cm	Computed: $350 - 2 \times 1$
Setup time, same pattern (reel reload)	5 min	Plant operating practice
Setup time, new pattern (full changeover)	15 min	Plant operating practice
Grammage of the analyzed orders	80 g/m ² (OFSET 1.HAMUR IVORY)	Production records
Number of slitting/cutting machines (plant total)	8	Public company information
Number of machines analyzed in this study	1 (Cameron slitter)	Data scope

5.3. Current Problems in the Process

Three interrelated problems were identified through inspection of the production records.

Mismatch between jumbo width and demand widths. The seven approved orders cover three finished widths (57, 64 and 70 cm) that do not combine into perfect 348 cm patterns. The trim loss per jumbo ranges from 6 cm to 28 cm in the data, equivalent to 1.72% to 8.05% of the net jumbo width. The weighted average is 4.50% of the total raw paper consumed.

Single-width pattern dominance. Two of the seven recorded orders use the single-width pattern 64-64-64-64-64, which leaves 28 cm of trim loss per jumbo. When orders are processed without cross-order width mixing, opportunities to combine compatible widths into tighter patterns are lost. The linear-programming reoptimization performed in Section 7 quantifies the gain that is mathematically achievable when the pattern set is redesigned.

Frequent pattern transitions. The seven orders use five distinct patterns. In the recorded arrival sequence the same pattern is never repeated back-to-back, so every order requires a full pattern changeover (15 min) rather than a simple reel reload (5 min). This sequencing is

suboptimal because two pairs of orders share identical patterns and could be processed consecutively.

6. Current System Analysis

This section quantifies the baseline performance of the Cameron slitter using the production data provided by the company. The analysis is restricted to the seven approved cutting plans contained in the data set so that every reported figure can be traced back to a specific record. No external assumptions about line speed, jumbo length or jumbo weight are introduced; all derived figures rely solely on the recorded width and kilogram values.

6.1. Cycle Time and Capacity Analysis

Each cutting cycle on the Cameron slitter consists of one productive run (the slitting of a jumbo) preceded by either a 5-minute reel reload (when the next jumbo uses the same pattern as the previous one) or a 15-minute full setup (when the pattern is changed). The productive runtime per jumbo cannot be expressed in absolute minutes from the available data because line speed and jumbo length are not recorded; however, the non-productive (setup) component is fully determined by the sequence of patterns and is therefore the variable used in the experimental analysis below.

Within the converting department, the eight installed machines collectively serve an annual plant capacity of 90,000 tons. The Cameron slitter is one of the primary slitters in this group. Because no per-machine throughput data are available, it is not possible to determine numerically whether the Cameron slitter is the local bottleneck. The qualitative bottleneck assessment is given in Section 6.2.

6.2. Bottleneck Analysis

Three potential constraint locations were considered: the upstream paper machine, the Cameron slitter and the downstream sheet-cutting operations. The paper machine produces continuously at the rated 90,000 ton/year capacity that defines the long-run output of the plant. The slitter operates in a make-to-order mode and is sensitive to the number of pattern transitions performed during a planning horizon. Each additional pattern transition adds 10 minutes of avoidable non-productive time relative to a same-pattern reel reload. The downstream sheet cutting operations are out of scope and are excluded from this analysis.

Because the long-run system output is governed by the paper machine, the slitter is not the steady-state bottleneck of the facility. However, the slitter can become a local bottleneck within the converting department whenever the planner chooses a strategy that produces many distinct cutting patterns. Section 7 demonstrates that this local bottleneck severity is controllable through the choice of pattern strategy and order grouping policy.

6.3. Trim Loss Analysis

Trim loss in the slitting operation is computed as the difference between the net jumbo width (348 cm) and the sum of finished-roll widths in the corresponding cutting pattern, divided by the net jumbo width. Table 5 reports the five distinct cutting patterns observed in the production sample, the trim loss in centimetres and percentage, and the kilograms of paper produced under each pattern. The kilogram of trim loss per pattern is computed using the proportional relation

$$\text{trim_loss_kg} = \text{produced_kg} \times \text{trim_cm} \div \text{used_widths_cm}$$

which holds because the finished rolls and the trim originate from the same jumbo length and grammage.

Table 5. Observed cutting patterns and computed trim loss.

Pattern (cm)	Used width (cm)	Trim (cm)	Trim (%)	Produced (kg)	Trim (kg)
70-70-70-70-57	337	11	3.16%	6,023.39	196.61
57-57-57-57-57-57	342	6	1.72%	11,527.76	202.24
64-64-64-64-64	320	28	8.05%	12,560.92	1,099.08
64-64-64-64-70	326	22	6.32%	12,833.91	866.09
70-70-70-64-64	338	10	2.87%	19,425.29	574.71
Total / Weighted Average	—	—	4.50%	62,371.26	2,938.74

The weighted-average trim loss is 4.50% of the total raw paper consumed. The minimum is 1.72% (the tightly packed 57-57-57-57-57-57 pattern, where $6 \times 57 = 342$ cm leaves only 6 cm of trim) and the maximum is 8.05% (the single-width 64-64-64-64-64 pattern, where $5 \times 64 =$

320 cm leaves 28 cm of trim because a sixth 64 cm strip would require 384 cm). This range demonstrates that pattern choice is the dominant lever for trim-loss control.

6.4. Setup Time Analysis

Setup time on the Cameron slitter takes one of two values depending on whether the next jumbo uses the same cutting pattern as the previous one. When the pattern is unchanged, the operator only reloads the reel, which takes approximately 5 minutes. When the pattern is changed, the slitter knives must be repositioned and the core shafts adjusted, which extends the setup to approximately 15 minutes. The total non-productive time over a sequence of orders is therefore

$$T_{setup} = 15 + (transitions \times 15) + (reloads \times 5)$$

where the leading 15 represents the initial setup before the first order, transitions counts the number of pattern changes within the sequence, and reloads counts the number of consecutive orders that share the same pattern. In the as-arrived sequence of the seven recorded orders no two consecutive orders share the same pattern, so transitions = 6 and reloads = 0, resulting in $T_{setup} = 15 + 90 + 0 = 105 \text{ minutes}$. This value is used as the baseline for the experimental analysis in Section 7.

7. Experimental Design / Scenario Analysis

This section applies a Design of Experiments (DoE) approach to evaluate how the choice of cutting pattern strategy and the choice of order sequencing jointly affect the performance of the Cameron slitter. Because direct experimentation on the production line is not feasible, the analysis uses the seven approved cutting plans from the data set as the demand sample and applies different decision rules to that sample. All response variables are derived from the recorded widths, kilograms and the two known setup-time values. The reoptimized pattern strategy is obtained from a linear programming (LP) model solved with Excel Solver; the same model and its results are documented in the supporting workbook.

7.1. Factors and Performance Measures

Two controllable factors were defined. Their levels are listed in Table 6.

Table 6. Factors and levels of the experimental design.

Factor	Levels	Description
A - Pattern strategy	Current / Reoptimized	Current keeps the five patterns recorded in the production data. Reoptimized uses the three LP-optimal patterns identified by an Excel Solver model that minimizes total raw-paper consumption subject to width-demand constraints. The active LP patterns are P1 (70-70-70-70-64), P2 (57-57-57-57-57-57) and P7 (70-64-64-64-64).
B - Order sequencing	As-arrived / Grouped	As-arrived processes the orders in the original SAP arrival sequence. Grouped reorders the same set of orders so that orders sharing a pattern run consecutively, converting full pattern changeovers (15 min) into reel reloads (5 min) where possible.

The full factorial design is $2 \times 2 = 4$ runs. Five response variables are recorded for each run:

- Y_1 — Trim loss (kg): kilograms of paper discarded as edge trim across the seven orders.
- Y_2 — Trim loss (%): trim loss as a fraction of total raw paper consumed.
- Y_3 — Material efficiency (%): 100 minus trim loss percentage.
- Y_4 — Number of pattern transitions in the sequence.
- Y_5 — Total setup time (minutes), computed from the formula in Section 6.4.

7.2. Designed Scenarios

The four scenarios were generated by combining the two levels of each factor. Run 1 reproduces the actual practice recorded in the data set and is therefore the baseline. Run 4 represents the joint application of both improvement levers and is therefore the proposed configuration. Runs 2 and 3 isolate the effect of each factor.

Table 7. Full factorial DoE results (4 runs).

#	Pattern strategy	Sequencing	Ptn.	Trn.	Setup (min)	Trim (kg)	Trim %	Eff. %
1	Current	As-arrived	5	6	105	2,938.74	4.50%	95.50%
2	Current	Grouped	5	4	85	2,938.74	4.50%	95.50%
3	Reoptimized	As-arrived	3	2	45	2,731.40	4.20%	95.80%
4	Reoptimized	Grouped	3	2	45	2,731.40	4.20%	95.80%

Note: Ptn. = number of distinct cutting patterns; Trn. = number of pattern transitions; Eff. = material efficiency.

7.3. Comparison of Results

The main effects of the two factors can be read directly from Table 7. Factor A (pattern strategy) is the only factor that affects trim loss: switching from the current pattern set to the LP-reoptimized one reduces trim loss from 2,938.74 kg to 2,731.40 kg over the seven-order sample, an absolute saving of 207.33 kg, equivalent to a 7.06% relative reduction. The trim-loss percentage falls from 4.50% to 4.20%, which is the mathematical lower bound for the analyzed demand pool as confirmed by the Excel Solver model. The reoptimized pattern set also reduces the number of distinct patterns from five to three.

Factor B (order sequencing) does not change the trim loss because the same patterns are used in both sequencing levels; it only affects the non-productive setup time. Under the current strategy, grouping orders by pattern reduces the number of pattern transitions from six to four, saving 20 minutes of setup time. Under the reoptimized strategy, however, only three distinct patterns remain, so the number of transitions is already at its minimum (two) regardless of the sequencing rule. The sequencing factor therefore becomes structurally inactive once the pattern set is sufficiently reduced — a finding that is itself an outcome of the experimental design.

The combined effect of moving from Run 1 (Current / As-arrived) to Run 4 (Reoptimized / Grouped) is a reduction of trim loss by 207.33 kg (−7.06%) and a reduction of total setup time by 60 minutes (−57.1%) on the analyzed sample. The number of pattern transitions falls from six to two. Table 8 summarizes the improvement potential.

Table 8. Summary of baseline versus proposed performance.

Performance measure	Baseline	Proposed	Improvement
Trim loss (kg)	2,938.74	2,731.40	−207.33 kg (−7.06%)
Trim loss (%)	4.50%	4.20%	−0.30 percentage points
Material efficiency (%)	95.50%	95.80%	+0.30 percentage points
Pattern transitions	6	2	−4 transitions
Setup time (min)	105	45	−60 min (−57.1%)

Note: Baseline corresponds to Run 1 (Current strategy / As-arrived sequencing); Proposed corresponds to Run 4 (Reoptimized strategy / Grouped sequencing).

An incidental but useful finding emerges from comparing Runs 3 and 4: the sequencing factor (B) ceases to influence the system once the pattern set is reoptimized. This occurs because the LP solution uses only three distinct patterns, which means the pattern-transition count is structurally bounded at two regardless of the order release sequence. The implication for the planner is that pattern reoptimization should be implemented first; once it is in place, the sequencing rule no longer needs separate management for setup-time control.

Projected onto the 90,000 ton/year plant capacity under the assumption that the Cameron slitter contributes proportionally to total output, the relative improvement in trim loss percentage corresponds to an annual reduction of approximately 274 tons of paper waste. This figure is intended only as an order-of-magnitude indicator because the per-machine output share of the Cameron slitter is not directly available.

8. Improvement Proposals

Two improvement proposals are derived from the experimental results. Each proposal is justified by the data and is feasible within the existing equipment and information infrastructure of the plant.

8.1. Cutting Pattern Reoptimization (LP-based)

The most impactful improvement is the systematic reassignment of customer demand to the LP-optimal cutting patterns. The DoE shows that this single change reduces trim loss by 7.06%

on the analyzed sample (from 2,938.74 kg to 2,731.40 kg) and reduces the number of distinct patterns from five to three. The recommended implementation is to introduce a planning aid that takes the open order book from SAP, generates the set of feasible patterns whose width sum is at least 90% of the net jumbo width, and solves a small linear program to minimize total raw-paper consumption. The output is a list of LP-optimal patterns with the corresponding jumbo allocations, which the planner can review before approving the production schedule. No capital investment is required because the procedure runs on top of the existing planning workflow and can be implemented in Excel using the Solver add-in.

An additional benefit of the LP-based approach is that the result is a mathematically defensible lower bound. The 4.20% trim percentage achieved by Run 4 is not merely an improvement over the current practice; it is the global minimum obtainable for the analyzed demand pool given the candidate pattern library. This provides the planning team with a clear performance benchmark against which future practice can be measured.

8.2. Order Grouping and Sequencing

The second proposal is a SAP planning rule that releases orders sharing the same cutting pattern consecutively rather than in arrival sequence. The DoE shows that this change reduces the number of pattern transitions from six to four under the current strategy, saving 20 minutes of setup time per planning cycle on the seven-order sample. Under the reoptimized strategy this saving is already captured by the reduction in the number of distinct patterns; the sequencing rule therefore has its largest incremental value during the transition period before pattern reoptimization is fully deployed. The proposed implementation is a sort criterion in the existing SAP planning view that uses the assigned cutting pattern as a secondary key after customer due dates. No master-data changes are required.

8.3. Combined Implementation and Expected Benefits

The two proposals are complementary because they address different sources of inefficiency. Pattern reoptimization reduces material loss; order sequencing reduces non-productive time. When implemented together they yield the Run 4 performance reported in Table 7: 7.06% lower trim loss and 57.1% lower setup time on the analyzed sample. Both proposals can be piloted on the Cameron slitter alone and, if successful, extended to the other primary slitters in the converting department.

In addition to the quantified material and time savings, the structured pattern-selection procedure provides three indirect benefits. First, it makes the planning logic transparent and reproducible, reducing dependence on individual planner experience. Second, it produces a smaller number of distinct patterns per planning cycle, which simplifies the work of the slitter operator. Third, it generates a baseline for continuous improvement: once the planning logic is automated, subsequent reductions in setup time (for example through SMED practices) can be measured against a stable reference.

9. Conclusion

This study analyzed the slitting operation on the Cameron slitter at the Alkim Kağıt Kemalpaşa facility through a combination of production-data review and experimental design. The analysis was restricted to the seven approved cutting plans contained in the available data set so that every reported figure could be traced back to a recorded value. The current cutting practice was found to generate a weighted-average trim loss of 4.50% on the analyzed sample, with a per-pattern range of 1.72% to 8.05%. The baseline non-productive setup time was 105 minutes for the seven-order sequence, of which 90 minutes were due to pattern transitions and 15 minutes to the initial setup.

A 2×2 full factorial Design of Experiments was applied to the data sample. The two factors were the pattern strategy (current or LP-reoptimized) and the order sequencing (as-arrived or grouped). The reoptimized pattern set was obtained by solving a linear-programming model with Excel Solver that minimizes total raw-paper consumption subject to width-demand constraints; the optimal solution uses only three patterns out of ten candidates. The combined Run 4 configuration reduces trim loss by 7.06% and setup time by 57.1% relative to the baseline. The 4.20% trim percentage achieved is the mathematical lower bound for the analyzed demand pool.

An additional structural finding emerges from the experiment: the order-sequencing factor becomes inactive once the pattern set is reduced to its LP optimum. This is because the resulting three-pattern schedule already operates at the minimum number of transitions. The implication for the planning team is that pattern reoptimization is a prerequisite improvement; sequencing rules should be tuned during the transition period and need no separate management once the LP-based planning aid is in place.

Two improvement proposals were derived from the analysis: an LP-based planning aid that automates the cutting-pattern reoptimization step, and a SAP sequencing rule that groups orders by their assigned pattern. Both proposals are implementable within the existing infrastructure of the plant and do not require capital investment. When applied together, the proposals are expected to reduce plant-level trim loss by approximately 274 tons per year, scaled from the sample-level result under the proportionality assumption stated in Section 7.3.

The main limitation of this study is the size of the production data sample, which restricts the statistical generalization of the results. The analysis is also restricted to the Cameron slitter and does not cover the downstream sheet-cutting operations, for which no measurement data were available. Future work should extend the analysis to a longer time horizon, include additional primary slitters, and validate the scenario-based improvements through a pilot run on the Cameron slitter before plant-wide deployment.

10. References

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11. Appendices

11.1. Appendix A. Excel Calculations

Figure A 1. . Cleaned raw production data used as the input for the Excel-based current system analysis.

Cleaned Raw Data				
Order ID	Pattern (Dizilim)	Width En (cm)	Trim Eni (cm)	Quantity (kg)
2000005777	70-70-70-70-57	57	11	1,018.793
2000005777	70-70-70-70-57	70	11	5,004.596
2000005778	57-57-57-57-57-57	57	6	4,009.656
2000005779	64-64-64-64-64	64	28	5,057.470
2000005780	64-64-64-64-70	70	22	2,755.747
2000005780	64-64-64-64-70	64	22	10,078.160
2000005781	64-64-64-64-64	64	28	7,503.450
2000005782	57-57-57-57-57-57	57	6	7,518.102
2000005786	70-70-70-64-64	70	10	12,068.967
2000005786	70-70-70-64-64	64	10	7,356.322

Figure A 2. Key operational parameters used in the Excel-based analysis.

Parameter	Value	Unit
Gross jumbo width	350	cm
Edge trim per side	1	cm
Net jumbo width	348	cm
Setup time, same pattern (reel reload)	5	min
Setup time, new pattern (full changeover)	15	min
Plant annual capacity	90,000	ton/year
Number of orders in DoE sample	7	—
Grammage of the analyzed orders	80	g/m ²

Figure A 3. Excel calculation of trim loss and material efficiency for the current cutting pattern set.

Pattern	S — Sum widths (cm)	T — Trim (cm)	Trim % per jumbo	Produced (kg)	Trim (kg)	Total raw (kg)	
70-70-70-70-57	337	11	3.1609%	6,023.3890	196.6091	6,219.9981	
57-57-57-57-57-57	342	6	1.7241%	11,527.7580	202.2414	11,729.9994	
64-64-64-64-64	320	28	8.0460%	12,560.9200	1,099.0805	13,660.0005	
64-64-64-64-70	326	22	6.3218%	12,833.9070	866.0919	13,699.9989	
70-70-70-64-64	338	10	2.8736%	19,425.2890	574.7127	20,000.0017	
TOTAL			4.4997%	62,371.2630	2,938.7356	65,309.9986	
Aggregate Metrics for the Current Pattern Set							
Total Produced (good output, kg)				62,371.2630			
Total Trim (kg)				2,938.7356			
Total raw paper consumed (kg)				65,309.9986			
Weighted-average Trim %				4.4997%			
Material efficiency %				95.5003%			
Min per-pattern Trim %				1.7241%			
Max per-pattern Trim %				8.0460%			

Figure A 4. Setup-time calculations for the four experimental runs under current and Solver-based sequencing strategies.

Run 1 — Current Strategy / As-arrived Sequencing					
7 orders processed in SAP arrival sequence (sorted by Sipariş ID).					
Position	Order/Block ID	Pattern	Same as previous ?	Setup contribution (min)	Cumulative (min)
1	5777	70-70-70-70-57	—	15	15
2	5778	57-57-57-57-57-57	No	15	30
3	5779	64-64-64-64-64	No	15	45
4	5780	64-64-64-64-70	No	15	60
5	5781	64-64-64-64-64	No	15	75
6	5782	57-57-57-57-57-57	No	15	90
7	5786	70-70-70-64-64	No	15	105
TOTAL			6	105	105 min
Run 2 — Current Strategy / Grouped Sequencing					
7 orders, same patterns; sequenced so identical patterns run consecutively.					
Position	Order/Block ID	Pattern	Same as previous ?	Setup contribution (min)	Cumulative (min)
1	5778	57-57-57-57-57-57	—	15	15
2	5782	57-57-57-57-57-57	Yes	5	20
3	5779	64-64-64-64-64	No	15	35
4	5781	64-64-64-64-64	Yes	5	40
5	5780	64-64-64-64-70	No	15	55
6	5786	70-70-70-64-64	No	15	70
7	5777	70-70-70-70-57	No	15	85
TOTAL			4	85	85 min
Run 3 — Solver-based Strategy / As-arrived Sequencing					
3 selected pattern blocks from the Solver model (P1, P2, P7), ordered by first appearance.					
Position	Order/Block ID	Pattern	Same as previous ?	Setup contribution (min)	Cumulative (min)
1	Block A	70-70-70-70-64	—	15	15
2	Block B	57-57-57-57-57-57	No	15	30
3	Block C	70-64-64-64-64	No	15	45
TOTAL			2	45	45 min
Run 4 — Solver-based Strategy / Grouped Sequencing					
3 selected pattern blocks; with only 3 distinct patterns, grouping does not further reduce the number of transitions.					
Position	Order/Block ID	Pattern	Same as previous ?	Setup contribution (min)	Cumulative (min)
1	Block A	70-70-70-70-64	—	15	15
2	Block B	57-57-57-57-57-57	No	15	30
3	Block C	70-64-64-64-64	No	15	45
TOTAL			2	45	45 min

Figure A 5. DoE factor structure, response variables, run results, and baseline–proposed improvement calculations.

Factor	Level 1	Level 2	What is varied				
A — Pattern strategy	Current	Reoptimized	Current = 5 patterns from production data. Reoptimized = 3 LP-optimal patterns from Solver: P1, P2, P7.				
B — Order sequencing	As-arrived	Grouped	As-arrived = SAP arrival order. Grouped = identical patterns run consecutively.				
Response Variables							
Symbol	Variable	Formula					
Y ₁	Trim loss (kg)	Total trim from active patterns					
Y ₂	Trim loss (%)	Total Trim / (Total Produced + Total Trim) * 100					
Y ₃	Material efficiency (%)	100 – Y ₂					
Y ₄	Pattern transitions (count)	k = number of i where pattern_i ≠ pattern_{i-1}					
Y ₅	Setup time (min)	T_setup = 15 + 15·k + 5·(n – 1 – k)					
DoE Run Results							
Run	Factor A	Factor B	Y ₁ Trim (kg)	Y ₂ Trim (%)	Y ₃ Eff. (%)	Y ₄ Trans.	Y ₅ Setup (min)
1	Current	As-arrived	2,938.74	4.50%	95.50%	6	105
2	Current	Grouped	2,938.74	4.50%	95.50%	4	85
3	Reoptimized	As-arrived	2,731.40	4.20%	95.80%	2	45
4	Reoptimized	Grouped	2,731.40	4.20%	95.80%	2	45
Baseline (Run 1) vs Proposed (Run 4) — Improvement Calculations							
Performance measure	Baseline (Run 1)	Proposed (Run 4)	Δ Absolute			Δ Relative (%)	
Trim loss (kg)	2,938.74	2,731.40	-207.33			-7.06%	
Trim loss (%)	4.50%	4.20%	-0.30%			-6.76%	
Material efficiency (%)	95.50%	95.80%	0.30%			0.32%	
Pattern transitions	6	2	-4			-66.67%	
Setup time (min)	105	45	-60			-57.14%	

Figure A 6. Main effects and interaction effects for trim loss (Y_1) and setup time (Y_3) based on the 2×2 factorial design.

Effect Table for Y_1 — Trim Loss (kg)					
	Factor B = As-arrived	Factor B = Grouped	Mean (row)		
Factor A = Current	2,938.74	2,938.74	2,938.74		
Factor A = Reoptimized	2,731.40	2,731.40	2,731.40		
Mean (col)	2,835.07	2,835.07			
Main effect of Factor A (Reoptim – Current)			-207.33		
Main effect of Factor B (Grouped – As-arrived)			0.00		
Interaction (A $\times$ B) effect = (R1 + R4 – R2 – R3) / 2			0.00		
Effect Table for Y_3 — Setup Time (min)					
	Factor B = As-arrived	Factor B = Grouped	Mean (row)		
Factor A = Current	105.00	85.00	95.00		
Factor A = Reoptimized	45.00	45.00	45.00		
Mean (col)	75.00	65.00			
Main effect of Factor A (Reoptim – Current)			-50.00		
Main effect of Factor B (Grouped – As-arrived)			-10.00		
Interaction (A $\times$ B) effect = (R1 + R4 – R2 – R3) / 2			10.00		

Figure A 7. Excel Solver model for LP-based cutting pattern optimization and constraint verification.

Demand Inputs							
Grammage (g/m ²)	80						
Width (cm)	Demand (kg)	Strip-meters needed					
57	12,546.551	275,143.66					
64	29,995.402	585,847.70					
70	19,829.310	354,094.82					
Candidate Pattern Library							
Pattern ID	Pattern	#57	#64	#70	Sum widths (cm)	Trim (cm)	Trim %
P1	70-70-70-70-64	0	1	4	344	4	1.15%
P2	57-57-57-57-57	6	0	0	342	6	1.72%
P3	70-70-70-64-64	0	2	3	338	10	2.87%
P4	70-70-70-70-57	1	0	4	337	11	3.16%
P5	70-70-64-64-64	0	3	2	332	16	4.60%
P6	70-70-70-64-57	1	1	3	331	17	4.89%
P7	70-64-64-64-64	0	4	1	326	22	6.32%
P8	70-70-64-64-57	1	2	2	325	23	6.61%
P9	70-70-70-57-57	2	0	3	324	24	6.90%
P10	64-64-64-64-64	0	5	0	320	28	8.05%
Decision Variables							
Pattern ID	x_p — Jumbo-meters used	Raw paper (kg)	Good output (kg)	Trim (kg)			
P1	55,368.77	15,414.67	15,237.49	177.18			
P2	45,857.28	12,766.67	12,546.55	220.11			
P3	0.00	0.00	0.00	0.00			
P4	0.00	0.00	0.00	0.00			
P5	0.00	0.00	0.00	0.00			
P6	0.00	0.00	0.00	0.00			
P7	132,619.73	36,921.33	34,587.23	2,334.11			
P8	0.00	0.00	0.00	0.00			
P9	0.00	0.00	0.00	0.00			
P10	0.00	0.00	0.00	0.00			
TOTAL	233,845.78	65,102.67	62,371.26	2,731.40			
Constraint Verification							
Width	Demand (strip-m)	Produced (strip-m)	Slack	Status			
57 cm	275,143.66	275,143.66	0.00	Satisfied			
64 cm	585,847.70	585,847.70	0.00	Satisfied			
70 cm	354,094.82	354,094.82	0.00	Satisfied			
Objective: Minimize Total Trim							
Total Trim to MINIMIZE (kg)				2,731.40 kg			
Total Raw Paper consumed (kg)				65,102.67 kg			
Trim % of raw paper				4.20%			

Figure A 8. Annualized trim loss and savings projection based on the baseline and proposed scenarios.

Quantity	Value	Unit
Plant annual capacity	90,000	ton/year
Baseline trim % (Run 1)	4.4997%	%
Proposed trim % (Run 4)	4.1955%	%
Annual baseline trim (ton)	4,049.7	ton/year
Annual proposed trim (ton)	3,776.0	ton/year
Annual savings (ton)	273.7	ton/year
Annual savings (%)	6.76%	%